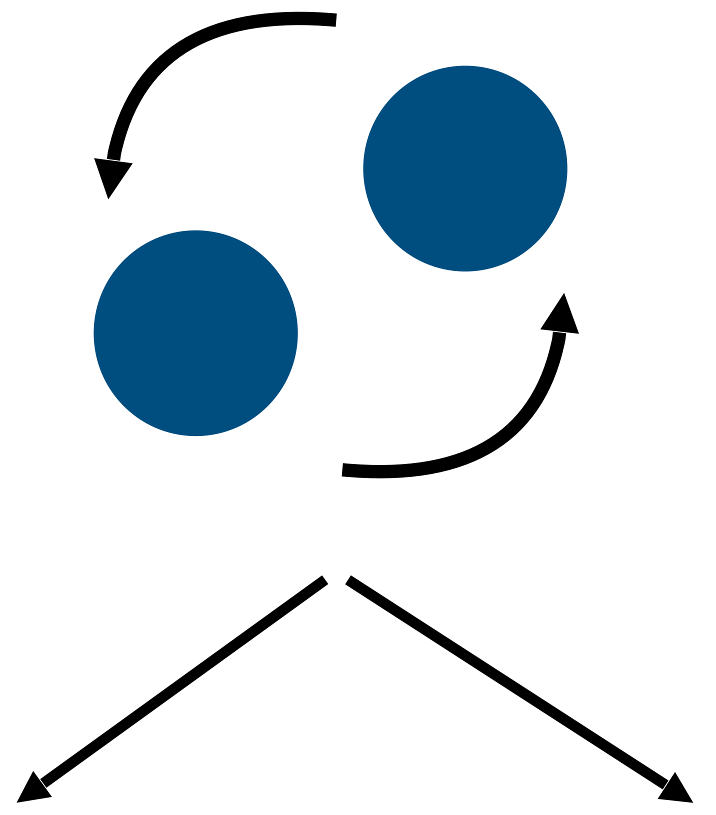
Moment scheme for neutrino transport in BNS mergers

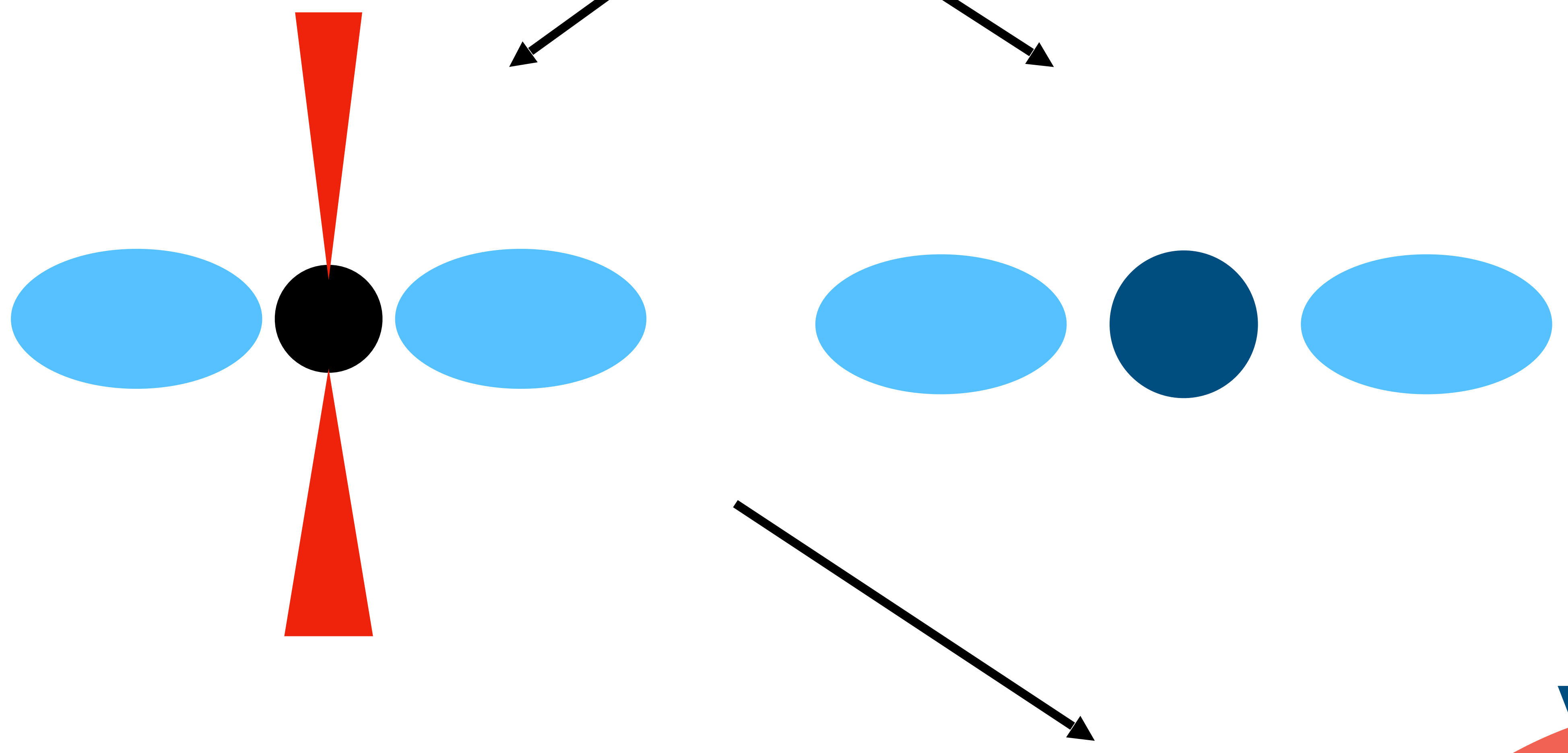
Carlo Musolino,
Based on CM, Rezzolla [arXiv:2304.09168]

Why neutrinos?

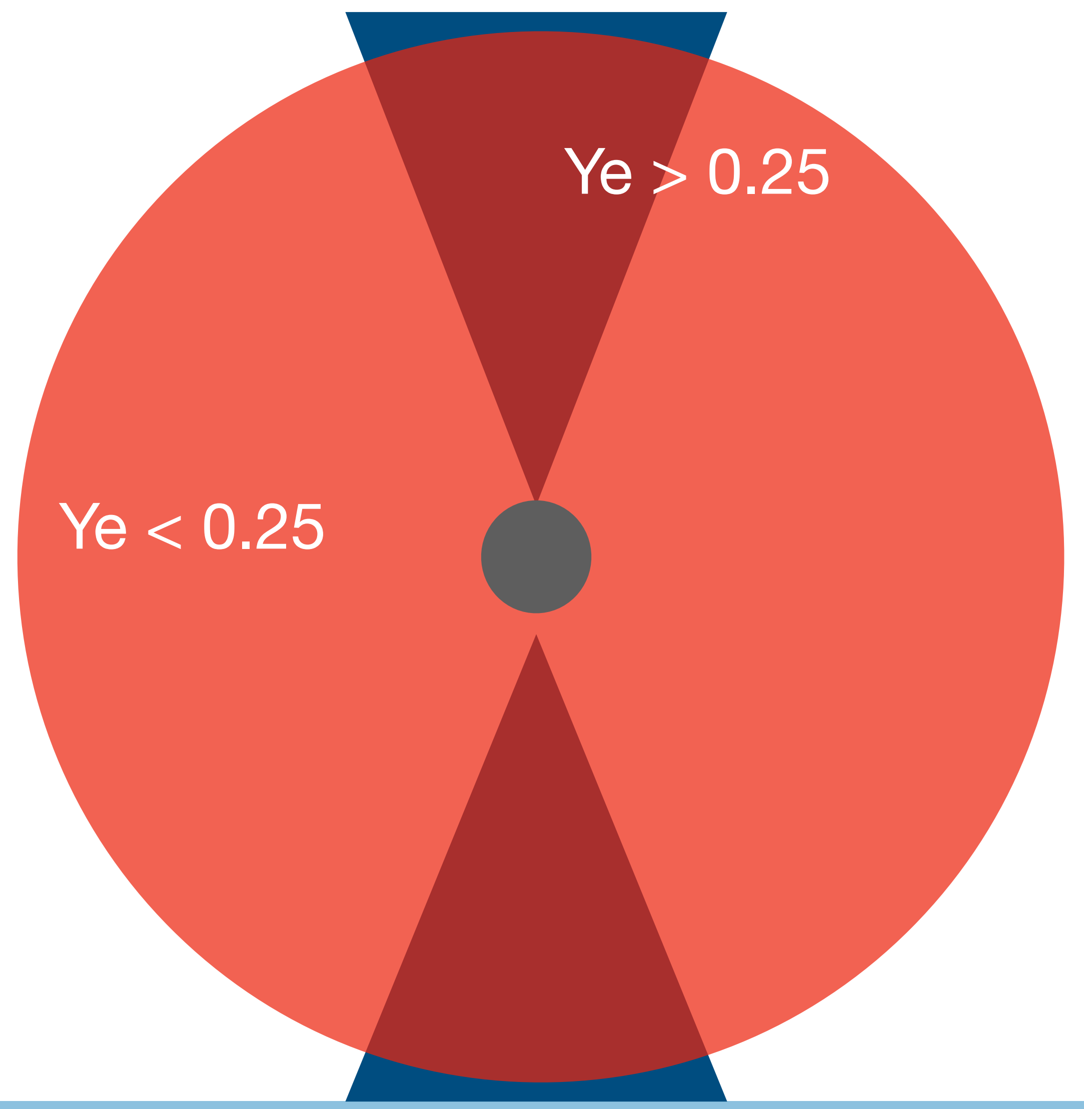
Inspiral: neutrinos are mostly irrelevant



Post-Merger: neutrino driven winds, disk cooling, reprocessing of the ejecta (increase Ye)



Expanding ejected material drives a shock in the ISM producing a Kilonova



How neutrinos?

The radiative transfer equation is a 6+1 dimensional PDE

Accounting for the effects of radiation in GRMHD simulations can be achieved with different strategies:

- **Leakage** type schemes account for local energy losses and composition changes from equilibrium rates directly [Ruffert+ arXiv:astro-ph/9509006 (Astronom. Atrophys.); Sekiguchi arXiv:1009.3358 (Class. And Quant. Grav.) and many more]

- **Moment-based schemes:** evolve a certain number of moments of the RTE (possibly averaged over the radiation energy) [Foucart+ **arXiv:1502.04146** (PRD); Weih+ **arXiv:2003.13580** (MNRAS); Radice, Bernuzzi, Perego **arXiv:2111.14858** (MNRAS); CM, Rezzolla **arXiv:2304.09168**]

- **Boltzmann solvers:** directly solve the RTE either with MC methods [Foucart+ **arXiv:1806.02349** (PRD)] , by direct discretization [Iwakami+ **DOI 10.3847/1538-4357/abb8c** (APJ)] or with Lattice Boltzmann techniques [Weih+ **arXiv:2007.05718** (MNRAS)]

Two-moment scheme

$$\partial_t \mathbf{U} + \partial_i \mathbf{F}^i(\mathbf{U}) = \mathbf{S}(\mathbf{U}) + \mathbf{C}(\mathbf{U})$$

M1 equations: can be written as a system of conservation laws.

$$\mathbf{U} = \sqrt{\gamma}\begin{pmatrix} \alpha f^0 n \\ E \\ F_i \end{pmatrix}$$

$$\mathbf{F^j} = \sqrt{\gamma}\begin{pmatrix} \alpha f^j n \\ \alpha F^j - \beta^j E \\ \alpha P^j_i - \beta^j F_i \end{pmatrix}$$

Evolved moments and fluxes

$$\mathbf{C} = \alpha\sqrt{\gamma}\begin{pmatrix} \eta^N - \kappa_a^N n \\ W\left\{\eta + \kappa_s J - \kappa\left(E - F^k v_k\right)\right\} \\ W\left(\eta - \kappa_a J\right) v_i - \kappa H_i \end{pmatrix}$$

Collisional source terms: coupling between radiation and fluid

$$\mathbf{S} = \sqrt{\gamma}\begin{pmatrix} 0 \\ \alpha P^{kj} K_{kj} - F^k \partial_k \alpha \\ F_j \partial_i \beta^j - E\partial_i \alpha + \frac{\alpha}{2} P^{jk} \partial_i \gamma_{jk} \end{pmatrix}$$

Geometric source terms

Two-moment scheme

Eulerian-frame moments: evolved variables

$$\mathbf{U} = \sqrt{\gamma}\begin{pmatrix} \alpha f^0 n \\ E \\ F_i \end{pmatrix}$$

$$\mathbf{F^j} = \sqrt{\gamma}\begin{pmatrix} \alpha f^j n \\ \alpha F^j - \beta^j E \\ \alpha P^j_i - \beta^j F_i \end{pmatrix}$$

Evolved moments and fluxes

$$\mathbf{C} = \alpha\sqrt{\gamma}\begin{pmatrix} \eta^N - \kappa^N_a\, n \\ W\left\{\, \eta + \kappa_s\, J - \kappa\left(E - F^k v_k\right)\,\right\} \\ W\left(\,\eta - \kappa_a J\right)\; v_i - \kappa\, H_i \end{pmatrix}$$

Collisional source terms: coupling between radiation and fluid

$$\mathbf{S} = \sqrt{\gamma}\begin{pmatrix} 0 \\ \alpha P^{kj} K_{kj} - F^k \partial_k \alpha \\ F_j \partial_i \beta^j - E \partial_i \alpha + \frac{\alpha}{2} P^{jk} \partial_i \gamma_{jk} \end{pmatrix}$$

Geometric source terms

Two-moment scheme

Fluid-frame moments

$$\mathbf{U} = \sqrt{\gamma}\begin{pmatrix}\alpha f^0 n \\ E \\ F_i\end{pmatrix}$$

$$\mathbf{F^j} = \sqrt{\gamma}\begin{pmatrix}\alpha f^j n \\ \alpha F^j - \beta^j E \\ \alpha P_i^j - \beta^j F_i\end{pmatrix}$$

Evolved moments and fluxes

$$\mathbf{C} = \alpha\sqrt{\gamma}\begin{pmatrix}\eta^N - \kappa_a^N\, n \\ W\left\{\eta + \kappa_s J - \kappa\left(E - F^k v_k\right)\right\} \\ W\left(\eta - \kappa_a J\right) v_i - \kappa H_i\end{pmatrix}$$

Collisional source terms: coupling between radiation and fluid

$$\mathbf{S} = \sqrt{\gamma}\begin{pmatrix}0 \\ \alpha\, P^{kj} K_{kj} - F^k \partial_k \alpha \\ F_j \partial_i \beta^j - E\partial_i \alpha + \frac{\alpha}{2} P^{jk} \partial_i \gamma_{jk}\end{pmatrix}$$

Geometric source terms

Two-moment scheme: closure

- In a moment scheme the analytic prescription for the higher-order moment fundamentally impacts the solution obtained by the scheme.

[Murchikova+ arXiv:1701.07027 (MNRAS)]

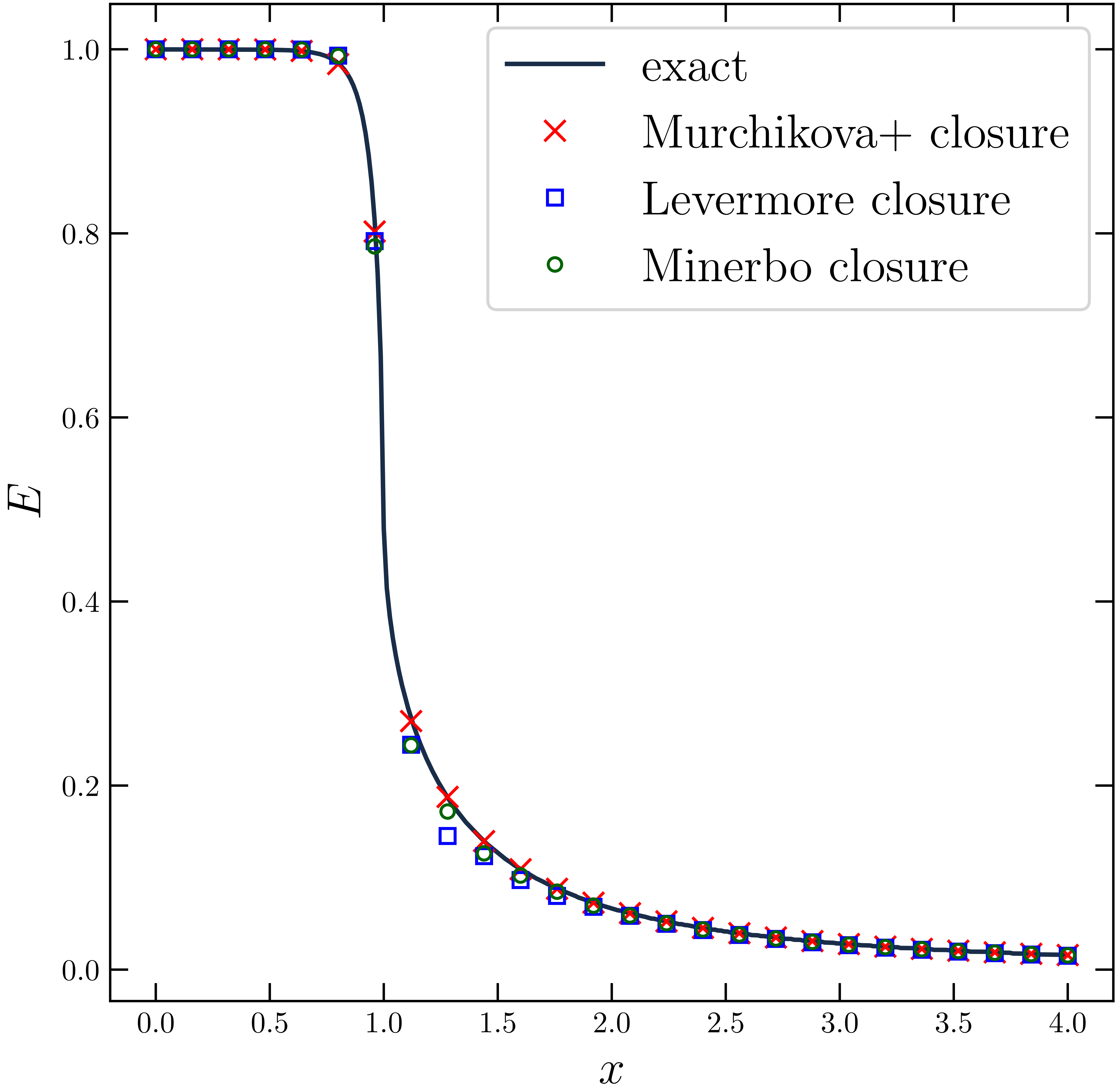
$$P^{ij} = d_{thin} P^{ij}_{thin} + d_{thick} P^{ij}_{thick}$$

$$d_{thin} = \frac{3}{2}\chi(\zeta) - \frac{1}{2}$$

$$d_{thick} = 1 - d_{thin}$$

$$\zeta = \sqrt{\frac{H^\mu H_\mu}{J^2}}$$

Figure from CM, Rezzolla arXiv:2304.09168



Two-moment scheme: fluxes discretization

Optically thick regime: RTE reduces to heat diffusion equation for energy density

$$\partial_t E - \frac{1}{3\kappa}\partial_x^2 E = \kappa_a (E_{\rm eq} - E)$$

Hyperbolic solver **introduces numerical dissipation** → Flux discretization needs to be **adapted to thermodynamic conditions**

First option: use the correct asymptotic flux in optically thick conditions [Foucart+ arXiv:1502.04146 (PRD)]

$$\mathcal{F}_{\rm corr} = A\mathcal{F} + (1-A)\mathcal{F}_{\rm asym}$$

$$\mathcal{F}_{\rm asym} = \sqrt{\gamma}\left(\frac{4}{3}W^2 \alpha v^d J_{\rm thick} - \beta^d E\right) - \alpha\frac{\sqrt{\gamma}W}{3\ \bar{\kappa}}\left(\gamma^{di} + v^d v^i\right)\partial_i J_{\rm thick}$$

Two-moment scheme: fluxes discretization

Optically thick regime: RTE reduces to heat diffusion equation for energy density

$$\partial_t E - \frac{1}{3\kappa}\partial_x^2 E = \kappa_a (E_{\rm eq} - E)$$

Hyperbolic solver **introduces numerical dissipation** → Flux discretization needs to be **adapted to thermodynamic conditions**

Second option: use a second order centered flux with a non-linear limiter (e.g. minmod) [Radice, Bernuzzi, Perego **arXiv:2111.14858** (MNRAS)]

$$\mathcal{F}_{\rm corr} = \mathcal{F}^{\rm HO} + A(1-\phi)\left(\mathcal{F}^{\rm HO} - \mathcal{F}^{\rm LO}\right)$$

Two-moment scheme: fluxes discretization

Optically thick regime: RTE reduces to heat diffusion equation for energy density

$$\partial_t E - \frac{1}{3\kappa}\partial_x^2 E = \kappa_a (E_{\rm eq} - E)$$

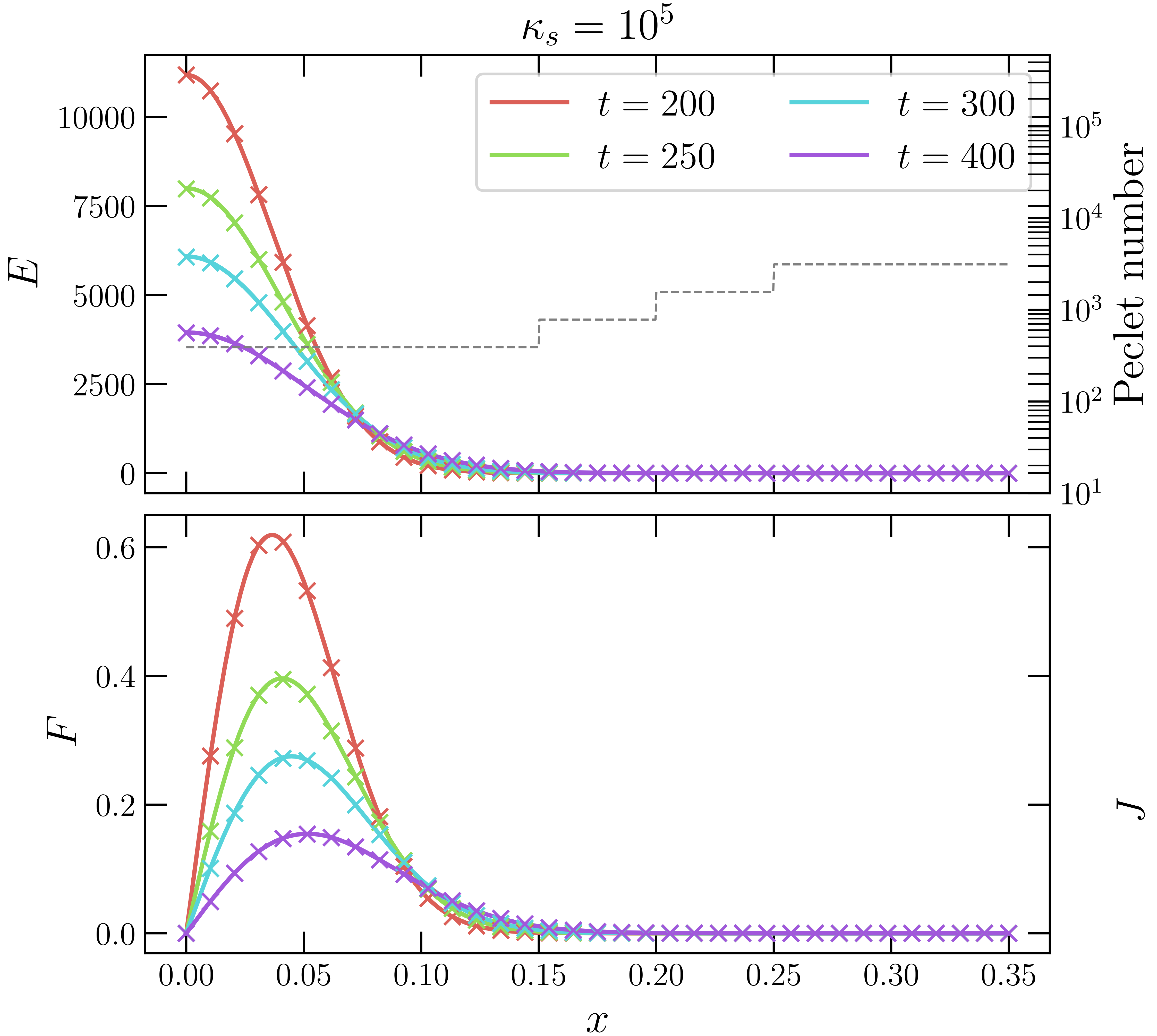
Hyperbolic solver **introduces numerical dissipation** → Flux discretization needs to be **adapted to thermodynamic conditions**

Third option: switch off diffusion in Riemann solver depending on the opacity

[Kuroda+ **arXiv:1501.06330** (ApJ); Skinner+ **arXiv:1806.07390** (ApJSS); CM, Rezzolla **arXiv:2304.09168**]

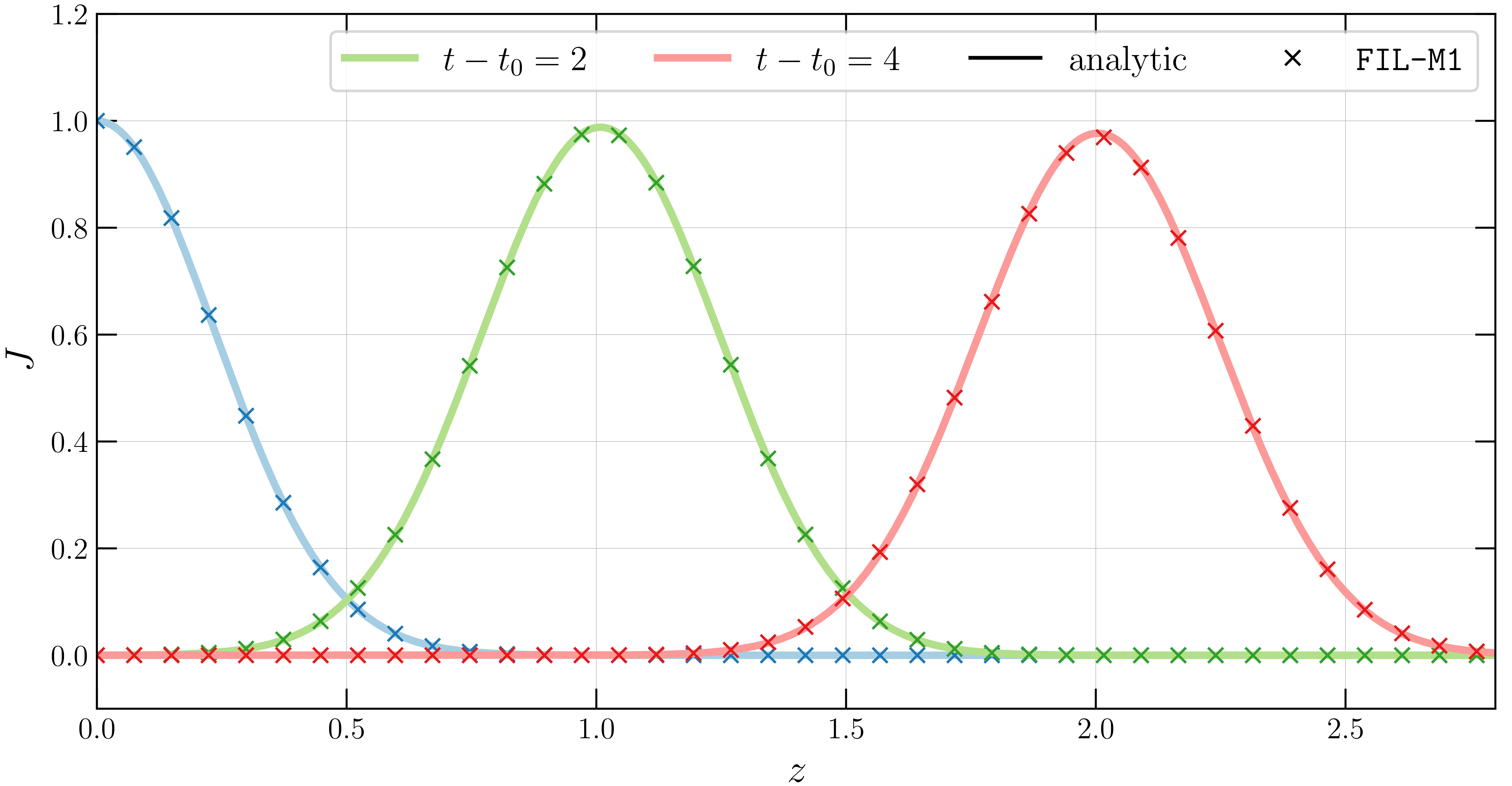
$$\mathcal{F}_{\rm corr} = \frac{c_{\rm R}\ F_{\rm L} - c_{\rm L}\ F_{\rm R} + A\ c_{\rm R}\ c_{\rm L}\ (U_{\rm R} + U_{\rm L})}{c_{\rm R} + c_{\rm L}}$$

Two-moment scheme: fluxes discretization



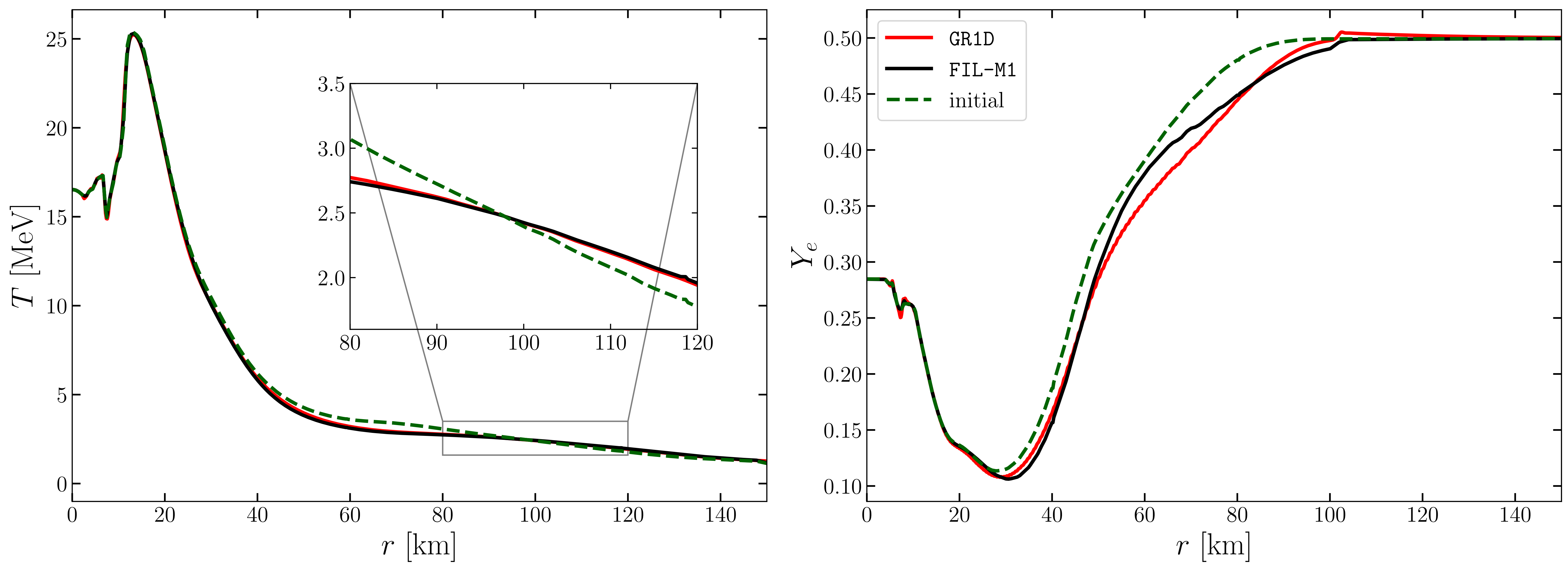
Figures from CM, Rezzolla arXiv:2304.09168

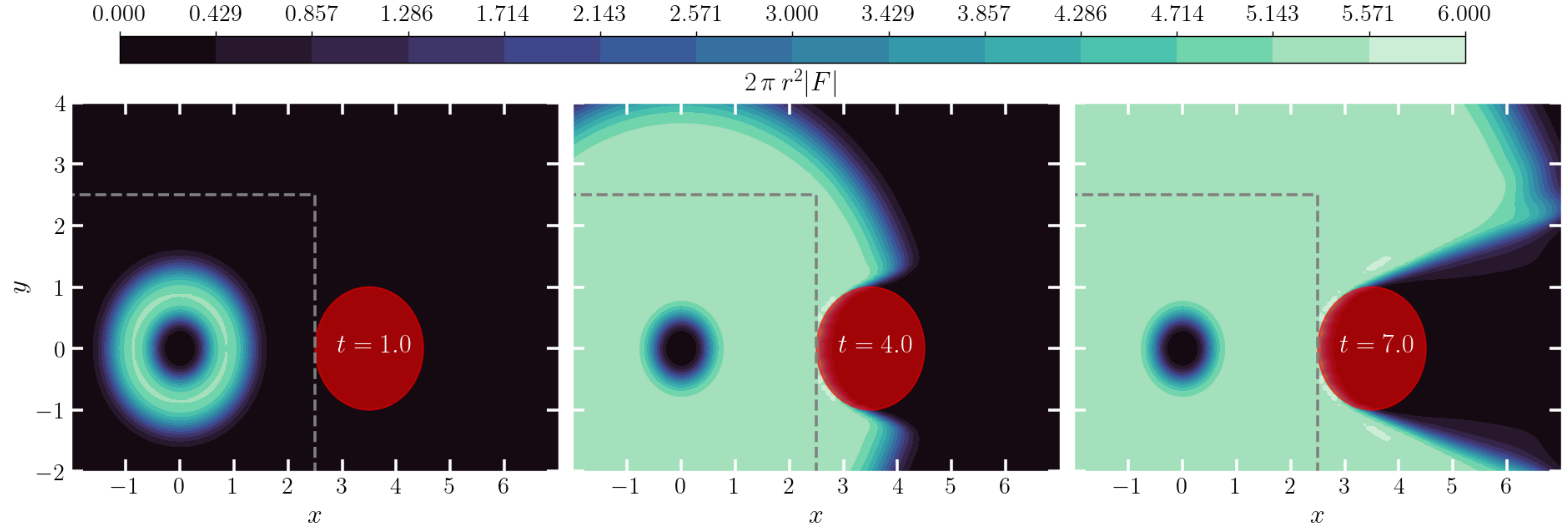
Radiation diffusion test: gaussian pulse on a stationary (left) and moving (bottom) medium with high scattering opacity



Two-moment scheme: collisional sources

- Collisional sources **provide the coupling** between radiation and the fluid





Figures from CM, Rezzolla arXiv:2304.09168

- Interaction terms in radiative transfer can become **stiff** and require implicit time-stepping and root-finding for evaluating the RHS

Application to BNSs

FIL-M1: M1 scheme implementation in the EinsteinToolkit [Löffler+ arXiv:1111.3344 (Class. Quant. Grav.)] coupled to the FIL GRMHD code [Most+ arXiv:1907.10328 (MNRAS)]

Evolution of **three effective neutrino species**

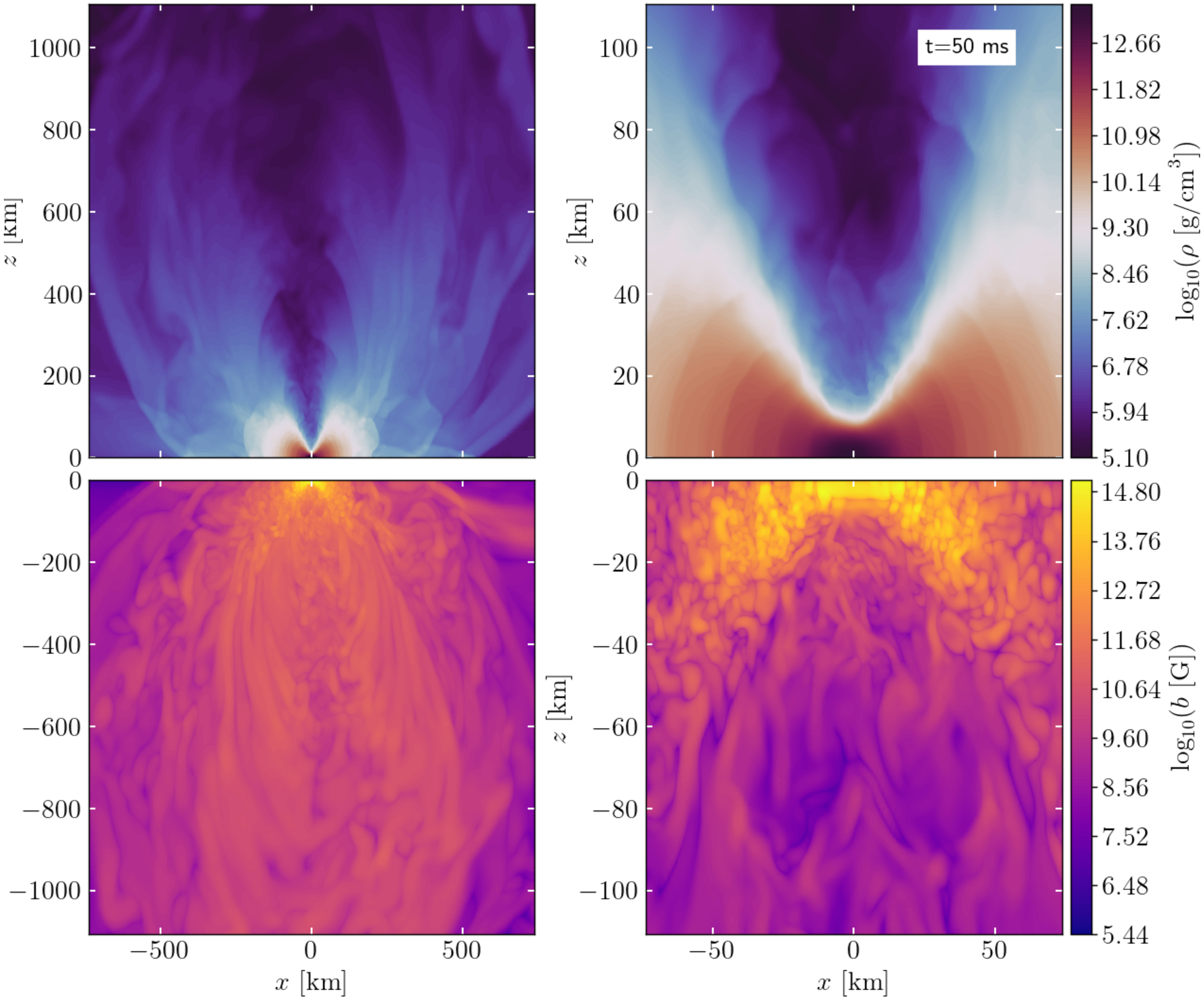
Energy averaged reaction rates include absorption, scattering and emission

Scattering	
Scattering on free nucleons	$\nu_i + N \to \nu_i + N\,,\ \bar{\nu}_i + N \to \bar{\nu}_i + N$
Scattering on heavy nuclei	$\nu_i + N \to \nu_i + N\,,\ \bar{\nu}_i + N \to \bar{\nu}_i + N$
Emission	
Absorption on free nucleons	$\nu_e + n \to p + e^-\,,\ \bar{\nu}_e + p \to n + e^+$
Emission	
Beta processes	$p + e^- \to \nu_e + n,\ n + e^+ \to \bar{\nu}_e + p$
Transverse plasmon decay	$\gamma \to \nu_x + \bar{\nu}_x$
Pair annihilation	$e^- + e^+ \to \nu_x + \bar{\nu}_x$
Nucleon-nucleon bremsstrahlung	$N + N \to \nu_x + \bar{\nu}_x$

Application to BNSs

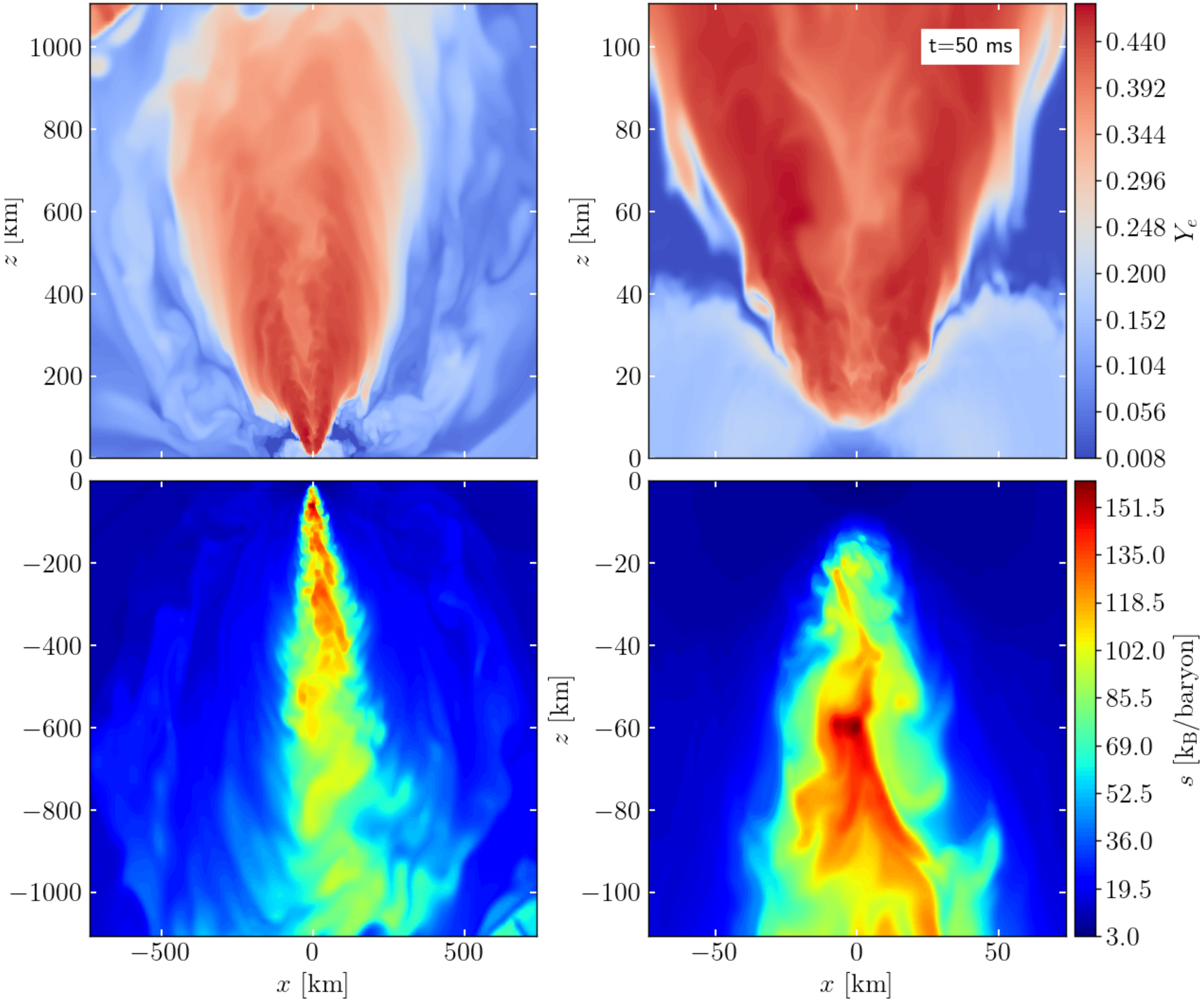
Magnetised BNS system described by the TNTyst equation of state.

M_1 $[M_\odot]$	M_2 $[M_\odot]$	q	EoS
1.18	1.57	0.75	TNTyst



Application to BNSs

Inclusion of neutrinos causes re-processing of the ejected material which becomes more electron rich.



Conclusions

- Self-consistent neutrino transport is an important ingredient for state-of-the art simulations of BNS mergers.

- We have presented a new implementation of a grey two-moment scheme for neutrino transport in BNS merger simulations.

- The M1 equations share many similarities with the GR-Hydro equations, however their numerical solution presents additional challenges which need to be properly addressed.

The closure

- Computing the closure requires finding the root of a scalar function: we do this with a standard NR scheme with Brent as a backup

$$R = \frac{H^\mu H_\mu - \zeta^2 J^2}{E^2}$$

$$\begin{aligned} H^\mu H_\mu = {} & H_0^2 + d_{thick}^2 H_{TT}^2 + \\ & d_{thin}^2 H_{tt}^2 + d_{thin} H_t^2 + \\ & d_{thick} H_T^2 + d_{thick} d_{thin} H_{tT}^2 \end{aligned}$$

$$\begin{aligned} \mathrm{b}_n &= W^2(E - 2Fv) + W(Fv - E) \\ b_n^t &= W^2(E - 2Fv) \\ b_v &= W^3(\; E - 2\, Fv) \\ b_v^t &= W^3 E \left(\frac{F^i v_i}{F} \right)^2 \\ b_v^T &= W \frac{W^2 - 1}{2W^2 + 1} \left(\; 4W^2\, Fv + (3 - 2W^2)E \right) \\ &+ \frac{W}{2W+1} \left[\; (2W^2 - 1)Fv + (3 - 2W^2)E \right] \\ b_F &= -W \\ b_F^t &= W\; E\; Fv \\ b_F^T &= W\, v^2 \end{aligned}$$

$$\begin{aligned} \mathrm{H}_0^2 &= b_v^2 v^i v_i + b_F^2 F_i F^i + 2 b_v b_F v^i F_i - b_n^2 \\ H_T^2 &= 2' \left(\; b_v b_v^t v^2 + (b_F^T)^2 F^2 + b_F^T b_v F v - b_n^T b_n \right) \\ H_t^2 &= 2 \left(\; b_v b_v^t v^2 + b_F b_f^t F + b_f^t b_v \frac{Fv}{F} + b_v^t b_F F v - b_n^t b_n \right) \\ H_{tt}^2 &= (b_v^t)^2 v^2 + (b_f^t)^2 F^2 + 2 b_F^T b_v^T F v - (b_n^T)^2 \\ H_{Tt}^2 &= 2 \left(\; b_v^t b_v^T v^2 + b_F^T b_f^t F + b_F^T b_v F v + b_f^t b_v^T \frac{Fv}{F} - b_n^T b_n^t \right) \end{aligned}$$

The transport rates

The microphysical rates included in the calculation of the collisional source terms are corrected according to three distinct procedures:

- To ensure the correct thermodynamical equilibrium, emissivities and opacities are recomputed from the Kirchoff theorem
- The temperature and composition in the black-body function is computed at the true beta-equilibrium for very optically thick media
- Opacities of processes which scale with the second power of the neutrino average energy are corrected with the improved information coming from the evolved moments.

$$\kappa^{\text{er,nr}}_{a,(\nu_x)} = \frac{Q^{\text{er,nr}}_{(\nu_x)}}{B^{\text{er,nr}}\left(T^*, \eta^*_{(\nu_x)}\right)} .$$

$$Q^{\text{er,nr}}_{(\nu_e,\bar{\nu}_e)} = B^{\text{er,nr}}\left(T^*, \eta^*_{(\nu_e,\bar{\nu}_e)}\right) \kappa^{\text{er,nr}}_{a,(\nu_e,\bar{\nu}_e)} ,$$

$$Y_l = Y_e^{\text{eq}} + Y_{\nu_e}\left(T^{\text{eq}}, Y_e^{\text{eq}}\right) - Y_{\bar{\nu}_e}\left(T^{\text{eq}}, Y_e^{\text{eq}}\right) ,$$

$$u = e\left(T^{\text{eq}}, Y_e^{\text{eq}}\right) + \frac{\rho}{m_b}\Big[Z_{\nu_e}\left(T^{\text{eq}}, Y_e^{\text{eq}}\right) + Z_{\bar{\nu}_e}\left(T^{\text{eq}}, Y_e^{\text{eq}}\right) + 4\, Z_{\nu_x}\left(T^{\text{eq}}, Y_e^{\text{eq}}\right)\Big] ,$$

$$\kappa^{\text{er,nr}}_{a,\,(\nu_e,\bar{\nu}_e)} = \kappa^{\text{er,nr}}_{a,\,(\nu_e,\bar{\nu}_e)}\Big|^{\text{eq}} \max\left(1, \left(\frac{T_{(\nu_e,\bar{\nu}_e)}}{T}\right)\right)^2 .$$

$$T_\nu := \frac{\mathcal{F}_2(\eta_\nu)}{\mathcal{F}_3(\eta_\nu)} \langle \epsilon_\nu \rangle ,$$